AI ASSISTANT LAB-16.3

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Task 1 – Hospital Management Database

Scenario :

Create schema and queries for a Hospital Management System

Prompt:

Create normalized schema for Hospital Management System
Tables: Doctors, Patients, Appointments
Include primary keys, foreign keys, constraints, valid date checks

Code :

sitantcoding > hospital_management_schema.sql

```sql
-- Hospital Management System (sqlite)
-- Normalized schema: Doctors, Patients, Appointments

BEGIN;

CREATE TABLE doctors (
    doctor_id BIGSERIAL PRIMARY KEY,    "BIGSERIAL": Unknown word.
    first_name VARCHAR(60) NOT NULL,    "VARCHAR": Unknown word.
    last_name VARCHAR(60) NOT NULL,    "VARCHAR": Unknown word.
    specialization VARCHAR(100) NOT NULL,    "VARCHAR": Unknown word.
    email VARCHAR(120) NOT NULL UNIQUE,    "VARCHAR": Unknown word.
    phone VARCHAR(20) NOT NULL UNIQUE,    "VARCHAR": Unknown word.
    license_number VARCHAR(40) NOT NULL UNIQUE,    "VARCHAR": Unknown word.
    years_experience INT NOT NULL DEFAULT 0,
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT chk_doctors_names_not_blank
        CHECK (BTRIM(first_name) <> '' AND BTRIM(last_name) <> ''),    "BTRIM": Unknown word.
    CONSTRAINT chk_doctors_specialization_not_blank
        CHECK (BTRIM(specialization) <> ''),    "BTRIM": Unknown word.
    CONSTRAINT chk_doctors_experience_non_negative
```

DBLEMS 23 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 23 SONARQUBE

```sql
 CONSTRAINT chk_doctors_experience_non_negative
        CHECK (years_experience >= 0)
);


CREATE TABLE patients (
   patient_id BIGSERIAL PRIMARY KEY,
   first_name VARCHAR(60) NOT NULL,
   last_name VARCHAR(60) NOT NULL,
   date_of_birth DATE NOT NULL,
```

```sql
    gender VARCHAR(20) NOT NULL,
    email VARCHAR(120) UNIQUE,
    phone VARCHAR(20) NOT NULL UNIQUE,
    blood_group VARCHAR(5),
    emergency_contact_name VARCHAR(120),
    emergency_contact_phone VARCHAR(20),
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT chk_patients_names_not_blank
        CHECK (BTRIM(first_name) <> '' AND BTRIM(last_name) <> ''),
    CONSTRAINT chk_patients_valid_dob
        CHECK (date_of_birth <= CURRENT_DATE),
    CONSTRAINT chk_patients_gender
        CHECK (gender IN ('Male', 'Female', 'Other')),
    CONSTRAINT chk_patients_blood_group
        CHECK (
            blood_group IS NULL
            OR blood_group IN ('A+', 'A-', 'B+', 'B-', 'AB+', 'AB-', 'O+', 'O-')
        )
);
```

Queries and output:

Query 1:

SELECT * FROM Appointments WHERE doctor_id = 1;

Output:

```
bash: /usr/share/bash-completion/bash_completion: No such file or directory
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$ sqlite3 -header -colum
n ./aiassitantcoding/hospital_management.db "SELECT * FROM appointments WHERE doctor_id = 1;"
appointment_id  doctor_id  patient_id  appointment_date  appointment_time  status     reason
       notes                           created_at          updated_at
--------------  ---------  ----------  ----------------  ----------------  ---------  ------------------
-----  ------------------------------  -------------------  -------------------
1               1          1           2026-03-19        10:00:00          Scheduled  Chest pain evaluat
ion    First consultation              2026-03-18 10:42:08  2026-03-18 10:42:08
3               1          3           2026-03-21        14:15:00          Scheduled  Routine cardiac ch
eckup  Family history of hypertension  2026-03-18 10:42:08  2026-03-18 10:42:08
```

Query 2

SELECT * FROM Appointments WHERE patient_id = 1;

Output:

```
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$ sqlite3 -header -column ./
aiassitantcoding/hospital_management.db "SELECT * FROM appointments WHERE patient_id = 1;"
appointment_id  doctor_id  patient_id  appointment_date  appointment_time  status     reason
 notes               created_at          updated_at
--------------  ---------  ----------  ----------------  ----------------  ---------  --------------------
 ------------------  ------------------  ------------------
1               1          1           2026-03-19        10:00:00          Scheduled  Chest pain evaluation
 First consultation  2026-03-18 10:42:08  2026-03-18 10:42:08
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$
```

Query 3:
SELECT doctor_id, COUNT(DISTINCT patient_id)
FROM Appointments
GROUP BY doctor_id;
Output:

```
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$ sqlite3 -header -column ./
aiassitantcoding/hospital_management.db "SELECT doctor_id, COUNT(DISTINCT patient_id) AS patient_count FROM
appointments GROUP BY doctor_id;"
doctor_id  patient_count
---------  -------------
1          2
2          1
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$
```

Justification:

Justification

- Copilot auto-generated **FK constraints**
- Ensured **3NF (no redundancy)**
- Used joins implicitly via FK

Task 2 – Real-Time Application: Online Shopping Database
Scenario:
Design a database for an E-commerce platform.
Prompt:

Create schema for e-commerce system
Tables: Users, Products, Orders, OrderDetails
Include relationships and normalization
Code

```sql
-- Task 2 - E-Commerce System (SQLite)
-- Normalized schema: users, products, orders, order_details

PRAGMA foreign_keys = ON;

BEGIN TRANSACTION;

CREATE TABLE users (
   user_id INTEGER PRIMARY KEY AUTOINCREMENT,
   full_name TEXT NOT NULL,
   email TEXT NOT NULL UNIQUE,
   phone TEXT UNIQUE,
   password_hash TEXT NOT NULL,
   is_active INTEGER NOT NULL DEFAULT 1 CHECK (is_active IN (0, 1)),
   created_at TEXT NOT NULL DEFAULT (datetime('now')),
   updated_at TEXT NOT NULL DEFAULT (datetime('now')),
   CHECK (trim(full_name) <> ''),
   CHECK (trim(email) <> ''),
   CHECK (instr(email, '@') > 1)
);

CREATE TABLE products (
   product_id INTEGER PRIMARY KEY AUTOINCREMENT,
   product_name TEXT NOT NULL,
   sku TEXT NOT NULL UNIQUE,
   description TEXT,
   unit_price NUMERIC NOT NULL,
   stock_quantity INTEGER NOT NULL DEFAULT 0,
   is_active INTEGER NOT NULL DEFAULT 1 CHECK (is_active IN (0, 1)),
   created_at TEXT NOT NULL DEFAULT (datetime('now')),
   updated_at TEXT NOT NULL DEFAULT (datetime('now')),
   CHECK (trim(product_name) <> ''),
   CHECK (trim(sku) <> ''),
   CHECK (unit_price >= 0),
```

```
   CHECK (stock_quantity >= 0)
);

CREATE TABLE orders (
   order_id INTEGER PRIMARY KEY AUTOINCREMENT,
   user_id INTEGER NOT NULL,
   order_date TEXT NOT NULL DEFAULT (datetime('now')),
   order_status TEXT NOT NULL DEFAULT 'Pending',
   payment_status TEXT NOT NULL DEFAULT 'Unpaid',
   shipping_address TEXT NOT NULL,
   order_total NUMERIC NOT NULL DEFAULT 0,
   created_at TEXT NOT NULL DEFAULT (datetime('now')),
   updated_at TEXT NOT NULL DEFAULT (datetime('now')),
   FOREIGN KEY (user_id)
       REFERENCES users (user_id)
       ON UPDATE CASCADE
       ON DELETE RESTRICT,
   CHECK (date(order_date) IS NOT NULL),
       CHECK  (order_status  IN  ('Pending',  'Confirmed',  'Shipped',  'Delivered',
'Cancelled')),
   CHECK (payment_status IN ('Unpaid', 'Paid', 'Refunded', 'Failed')),
   CHECK (trim(shipping_address) <> ''),
   CHECK (order_total >= 0)
);

CREATE TABLE order_details (
   order_detail_id INTEGER PRIMARY KEY AUTOINCREMENT,
   order_id INTEGER NOT NULL,
   product_id INTEGER NOT NULL,
   quantity INTEGER NOT NULL,
   unit_price NUMERIC NOT NULL,
   line_total NUMERIC NOT NULL,
   created_at TEXT NOT NULL DEFAULT (datetime('now')),
   updated_at TEXT NOT NULL DEFAULT (datetime('now')),
   FOREIGN KEY (order_id)
       REFERENCES orders (order_id)
       ON UPDATE CASCADE
       ON DELETE CASCADE,
   FOREIGN KEY (product_id)
       REFERENCES products (product_id)
```

```sql
        ON UPDATE CASCADE
        ON DELETE RESTRICT,
    CHECK (quantity > 0),
    CHECK (unit_price >= 0),
    CHECK (line_total >= 0),
    UNIQUE (order_id, product_id)
);

CREATE INDEX idx_orders_user_id
    ON orders (user_id);

CREATE INDEX idx_orders_order_date
    ON orders (order_date);

CREATE INDEX idx_order_details_order_id
    ON order_details (order_id);

CREATE INDEX idx_order_details_product_id
    ON order_details (product_id);

COMMIT;
```

Queries and output

```
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$ sqlite3 -header -column ./
aiassitantcoding/ecommerce.db "SELECT * FROM orders WHERE user_id = 1;"
order_id  user_id  order_date          order_status  payment_status  shipping_address      order_total  cre
ated_at           updated_at
--------  -------  ------------------  ------------  --------------  --------------------  -----------  ---
----------------  ------------------
1         1        2026-03-05 10:15:00  Delivered     Paid            Bengaluru, Karnataka  3497         202
6-03-18 11:02:46  2026-03-18 11:02:46
2         1        2026-03-12 16:45:00  Shipped       Paid            Bengaluru, Karnataka  3798         202
6-03-18 11:02:46  2026-03-18 11:02:46
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$ sqlite3 -header -column ./
aiassitantcoding/ecommerce.db "SELECT product_id, SUM(quantity) AS total FROM order_details GROUP BY product
_id ORDER BY total DESC LIMIT 1;"
product_id  total
----------  -----
1           3
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$ sqlite3 -header -column ./
aiassitantcoding/ecommerce.db "SELECT SUM(p.unit_price * od.quantity) AS revenue FROM orders o JOIN order_de
tails od ON o.order_id = od.order_id JOIN products p ON p.product_id = od.product_id WHERE strftime('%m', o.
order_date) = '03';"
revenue
-------
10593
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$
```

Task 3 – Library Database

Task:
Design schema for a Library Management System.
Prompt:
Create library schema
Tables: Books, Members, Loans
Suggest indexes for performance
Code

```
- Task - Library Management System (SQLite)
-- Schema + sample data

PRAGMA foreign_keys = ON;

BEGIN TRANSACTION;

CREATE TABLE books (
   book_id INTEGER PRIMARY KEY AUTOINCREMENT,
   isbn TEXT NOT NULL UNIQUE,
   title TEXT NOT NULL,
   author_name TEXT NOT NULL,
   category TEXT,
   published_year INTEGER,
   total_copies INTEGER NOT NULL DEFAULT 1,
   available_copies INTEGER NOT NULL DEFAULT 1,
   shelf_code TEXT,
   created_at TEXT NOT NULL DEFAULT (datetime('now')),
   updated_at TEXT NOT NULL DEFAULT (datetime('now')),
   CHECK (trim(isbn) <> ''),
   CHECK (trim(title) <> ''),
   CHECK (trim(author_name) <> ''),
   CHECK (published_year IS NULL OR published_year BETWEEN 1450 AND 2100),
   CHECK (total_copies >= 0),
   CHECK (available_copies >= 0),
   CHECK (available_copies <= total_copies)
);

CREATE TABLE members (
   member_id INTEGER PRIMARY KEY AUTOINCREMENT,
```

```sql
    full_name TEXT NOT NULL,
    email TEXT NOT NULL UNIQUE,
    phone TEXT UNIQUE,
    membership_date TEXT NOT NULL DEFAULT (date('now')),
    status TEXT NOT NULL DEFAULT 'Active',
    created_at TEXT NOT NULL DEFAULT (datetime('now')),
    updated_at TEXT NOT NULL DEFAULT (datetime('now')),
    CHECK (trim(full_name) <> ''),
    CHECK (instr(email, '@') > 1),
    CHECK (status IN ('Active', 'Suspended', 'Inactive')),
    CHECK (date(membership_date) IS NOT NULL)
);

CREATE TABLE loans (
    loan_id INTEGER PRIMARY KEY AUTOINCREMENT,
    book_id INTEGER NOT NULL,
    member_id INTEGER NOT NULL,
    loan_date TEXT NOT NULL,
    due_date TEXT NOT NULL,
    return_date TEXT,
    loan_status TEXT NOT NULL DEFAULT 'Borrowed',
    fine_amount NUMERIC NOT NULL DEFAULT 0,
    created_at TEXT NOT NULL DEFAULT (datetime('now')),
    updated_at TEXT NOT NULL DEFAULT (datetime('now')),
    FOREIGN KEY (book_id)
        REFERENCES books (book_id)
        ON UPDATE CASCADE
        ON DELETE RESTRICT,
    FOREIGN KEY (member_id)
        REFERENCES members (member_id)
        ON UPDATE CASCADE
        ON DELETE RESTRICT,
    CHECK (date(loan_date) IS NOT NULL),
    CHECK (date(due_date) IS NOT NULL),
    CHECK (return_date IS NULL OR date(return_date) IS NOT NULL),
    CHECK (date(due_date) >= date(loan_date)),
    CHECK (return_date IS NULL OR date(return_date) >= date(loan_date)),
    CHECK (loan_status IN ('Borrowed', 'Returned', 'Overdue', 'Lost')),
    CHECK (fine_amount >= 0)
);
```

```sql
-- Suggested performance indexes
CREATE INDEX idx_books_title
   ON books (title);

CREATE INDEX idx_books_author
   ON books (author_name);

CREATE INDEX idx_members_email
   ON members (email);

CREATE INDEX idx_loans_member_status
   ON loans (member_id, loan_status);

CREATE INDEX idx_loans_book_status
   ON loans (book_id, loan_status);

CREATE INDEX idx_loans_due_date
   ON loans (due_date);

-- Sample books
INSERT OR IGNORE INTO books
(isbn, title, author_name, category, published_year, total_copies, available_copies,
shelf_code)
VALUES
('9780131103627', 'The C Programming Language', 'Brian W. Kernighan', 'Programming',
1988, 5, 3, 'CS-A1'),
('9780132350884', 'Clean Code', 'Robert C. Martin', 'Software Engineering', 2008, 4,
2, 'CS-B2'),
('9780262033848', 'Introduction to Algorithms', 'Thomas H. Cormen', 'Algorithms',
2009, 3, 1, 'CS-C3'),
('9780596009205', 'Head First Java', 'Kathy Sierra', 'Programming', 2005, 6, 4,
'CS-D4');

-- Sample members
INSERT OR IGNORE INTO members
(full_name, email, phone, membership_date, status)
VALUES
('Riya Verma', 'riya.verma@example.com', '9100000001', '2026-01-10', 'Active'),
('Arun Patel', 'arun.patel@example.com', '9100000002', '2026-02-04', 'Active'),
```

```
('Sneha Nair', 'sneha.nair@example.com', '9100000003', '2026-02-15', 'Active');

-- Sample loans
INSERT OR IGNORE INTO loans
(book_id, member_id, loan_date, due_date, return_date, loan_status, fine_amount)
VALUES
(1, 1, '2026-03-10', '2026-03-20', NULL, 'Borrowed', 0),
(2, 1, '2026-03-01', '2026-03-10', '2026-03-09', 'Returned', 0),
(3, 2, '2026-03-05', '2026-03-15', NULL, 'Borrowed', 0),
(4, 3, '2026-02-20', '2026-03-01', '2026-03-03', 'Returned', 20);

COMMIT;
```

Queries and output:

```
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$ sqlite3 -header -column ./aiassitantc
oding/library.db "SELECT * FROM loans;"
loan_id  book_id  member_id  loan_date   due_date    return_date  loan_status  fine_amount  created_at          update
d_at
-------  -------  ---------  ----------  ----------  -----------  -----------  -----------  ------------------  ------
-------------
1        1        1          2026-03-10  2026-03-20               Borrowed     0            2026-03-18 11:31:00  2026-0
3-18 11:31:00
2        2        1          2026-03-01  2026-03-10  2026-03-09   Returned     0            2026-03-18 11:31:00  2026-0
3-18 11:31:00
3        3        2          2026-03-05  2026-03-15               Borrowed     0            2026-03-18 11:31:00  2026-0
3-18 11:31:00
4        4        3          2026-02-20  2026-03-01  2026-03-03   Returned     20           2026-03-18 11:31:00  2026-0
3-18 11:31:00
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$ sqlite3 -header -column ./aiassitantc
oding/library.db "SELECT * FROM loans WHERE date(loan_date) < date('now', '-30 day');"
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$ sqlite3 -header -column ./aiassitantc
oding/library.db "SELECT * FROM loans WHERE date(loan_date) < date('now', '-30 day');"
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$ sqlite3 -header -column ./aiassitantc
oding/library.db "SELECT member_id, COUNT(*) AS loan_count FROM loans GROUP BY member_id;"
member_id  loan_count
---------  ----------
1          2
2          1
3          1
```

Justification

Index improves **search speed**

Copilot suggested indexing on **date column**

Task 4: Optimize Queries

Instructions:

- Use AI tools to analyze query efficiency.

• Optimize inefficient queries using joins, indexes, or reducing subqueries.
• Test optimized queries for correctness.
Starter Code Example:
-- Original query
SELECT * FROM Books WHERE author_id IN (SELECT author_id FROM
Authors WHERE country='UK');
-- Optimized query
SELECT b.*
FROM Books b
JOIN Authors a ON b.author_id = a.author_id
WHERE a.country = 'UK';
Prompt : ptimize query using JOIN instead of subquery
Code
-- Original
SELECT * FROM Books WHERE author_id IN
(SELECT author_id FROM Authors WHERE country='UK');

-- Optimized
SELECT b.*
FROM Books b
JOIN Authors a ON b.author_id = a.author_id
WHERE a.country='UK';

```
-- Original
SELECT * FROM Books WHERE author_id IN
(SELECT author_id FROM Authors WHERE country='UK');

-- Optimized
SELECT b.*
```

```
FROM Books b
JOIN Authors a ON b.author_id = a.author_id
WHERE a.country='UK';
```

Output:

```
book_id | author_id | title                                | category
------- | --------- | ------------------------------------ | ----------
101     | 1         | 1984                                 | Fiction
102     | 2         | Harry Potter and the Philosopher's Stone| Fantasy
```

Justification

JOIN reduces **nested query overhead**

Faster on large datasets

Task 5: University Course Registration

Scenario:

SR University needs a course registration system for students enrolling in

different courses under faculty supervision.

• Use Copilot to design schema tables: Students, Courses, Faculty, and Registrations.

• Define relationships:

• Each student can register for multiple courses.

• Each course is taught by a faculty member.

• Write SQL queries to:

• List all students enrolled in a specific course.

• Find faculty members teaching more than 2 courses.

• Retrieve courses with the highest number of registrations

Prompt:

ceate schema for Students, Courses, Faculty, Registrations

Define relationships and constraints

Code

```
- Task 5 - University Course Registration (SQLite)
-- Schema + sample data + required queries

PRAGMA foreign_keys = ON;

BEGIN TRANSACTION;

CREATE TABLE faculty (
   faculty_id INTEGER PRIMARY KEY AUTOINCREMENT,
   full_name TEXT NOT NULL,
   email TEXT NOT NULL UNIQUE,
   department TEXT NOT NULL,
   designation TEXT,
   hire_date TEXT,
   is_active INTEGER NOT NULL DEFAULT 1 CHECK (is_active IN (0, 1)),
   created_at TEXT NOT NULL DEFAULT (datetime('now')),
   updated_at TEXT NOT NULL DEFAULT (datetime('now')),
   CHECK (trim(full_name) <> ''),
   CHECK (instr(email, '@') > 1),
   CHECK (trim(department) <> ''),
   CHECK (hire_date IS NULL OR date(hire_date) IS NOT NULL)
);

CREATE TABLE students (
   student_id INTEGER PRIMARY KEY AUTOINCREMENT,
   roll_no TEXT NOT NULL UNIQUE,
   full_name TEXT NOT NULL,
   email TEXT NOT NULL UNIQUE,
   phone TEXT UNIQUE,
   program TEXT NOT NULL,
   semester INTEGER NOT NULL,
   enrollment_date TEXT NOT NULL DEFAULT (date('now')),
   status TEXT NOT NULL DEFAULT 'Active',
   created_at TEXT NOT NULL DEFAULT (datetime('now')),
   updated_at TEXT NOT NULL DEFAULT (datetime('now')),
   CHECK (trim(roll_no) <> ''),
   CHECK (trim(full_name) <> ''),
   CHECK (instr(email, '@') > 1),
   CHECK (trim(program) <> ''),
```

```sql
    CHECK (semester BETWEEN 1 AND 12),
    CHECK (date(enrollment_date) IS NOT NULL),
    CHECK (status IN ('Active', 'Graduated', 'Dropped'))
);

CREATE TABLE courses (
    course_id INTEGER PRIMARY KEY AUTOINCREMENT,
    course_code TEXT NOT NULL UNIQUE,
    course_title TEXT NOT NULL,
    credits INTEGER NOT NULL,
    department TEXT NOT NULL,
    faculty_id INTEGER NOT NULL,
    max_capacity INTEGER NOT NULL DEFAULT 60,
    created_at TEXT NOT NULL DEFAULT (datetime('now')),
    updated_at TEXT NOT NULL DEFAULT (datetime('now')),
    FOREIGN KEY (faculty_id)
        REFERENCES faculty (faculty_id)
        ON UPDATE CASCADE
        ON DELETE RESTRICT,
    CHECK (trim(course_code) <> ''),
    CHECK (trim(course_title) <> ''),
    CHECK (trim(department) <> ''),
    CHECK (credits BETWEEN 1 AND 6),
    CHECK (max_capacity > 0)
);

CREATE TABLE registrations (
    registration_id INTEGER PRIMARY KEY AUTOINCREMENT,
    student_id INTEGER NOT NULL,
    course_id INTEGER NOT NULL,
    academic_year TEXT NOT NULL,
    semester_term TEXT NOT NULL,
    registration_date TEXT NOT NULL DEFAULT (date('now')),
    grade TEXT,
    status TEXT NOT NULL DEFAULT 'Enrolled',
    created_at TEXT NOT NULL DEFAULT (datetime('now')),
    updated_at TEXT NOT NULL DEFAULT (datetime('now')),
    FOREIGN KEY (student_id)
        REFERENCES students (student_id)
        ON UPDATE CASCADE
```

```sql
        ON DELETE CASCADE,
    FOREIGN KEY (course_id)
        REFERENCES courses (course_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CHECK (academic_year GLOB '[0-9][0-9][0-9][0-9]-[0-9][0-9][0-9][0-9]'),
    CHECK (semester_term IN ('Odd', 'Even', 'Summer')),
    CHECK (date(registration_date) IS NOT NULL),
    CHECK (grade IS NULL OR grade IN ('O', 'A+', 'A', 'B+', 'B', 'C', 'P', 'F')),
    CHECK (status IN ('Enrolled', 'Completed', 'Dropped', 'Withdrawn')),
    UNIQUE (student_id, course_id, academic_year, semester_term)
);

-- Suggested performance indexes
CREATE INDEX idx_students_program_semester
    ON students (program, semester);

CREATE INDEX idx_courses_faculty
    ON courses (faculty_id);

CREATE INDEX idx_registrations_course
    ON registrations (course_id);

CREATE INDEX idx_registrations_student
    ON registrations (student_id);

CREATE INDEX idx_registrations_course_status
    ON registrations (course_id, status);

-- Sample faculty
INSERT OR IGNORE INTO faculty (full_name, email, department, designation, hire_date,
is_active)
VALUES
('Dr. Anil Kumar', 'anil.kumar@sru.edu', 'Computer Science', 'Professor',
'2015-06-10', 1),
('Dr. Priya Menon', 'priya.menon@sru.edu', 'Computer Science', 'Associate Professor',
'2018-01-12', 1),
('Dr. Rakesh Iyer', 'rakesh.iyer@sru.edu', 'Electronics', 'Assistant Professor',
'2020-08-01', 1);
```

```sql
-- Sample students
INSERT OR IGNORE INTO students (roll_no, full_name, email, phone, program, semester,
enrollment_date, status)
VALUES
('SRU23CSE001', 'Aman Verma', 'aman.verma@sru.edu', '9800000001', 'B.Tech CSE', 5,
'2023-08-01', 'Active'),
('SRU23CSE002', 'Nisha Rao', 'nisha.rao@sru.edu', '9800000002', 'B.Tech CSE', 5,
'2023-08-01', 'Active'),
('SRU22ECE010', 'Rahul Nair', 'rahul.nair@sru.edu', '9800000010', 'B.Tech ECE', 7,
'2022-08-01', 'Active'),
('SRU24CSE005', 'Megha Shah', 'megha.shah@sru.edu', '9800000005', 'B.Tech CSE', 3,
'2024-08-01', 'Active');

-- Sample courses
INSERT OR IGNORE INTO courses (course_code, course_title, credits, department,
faculty_id, max_capacity)
VALUES
('CS301', 'Database Management Systems', 4, 'Computer Science', 1, 70),
('CS302', 'Operating Systems', 4, 'Computer Science', 1, 65),
('CS303', 'Design and Analysis of Algorithms', 4, 'Computer Science', 1, 60),
('CS304', 'Computer Networks', 3, 'Computer Science', 2, 60),
('EC401', 'Digital Signal Processing', 4, 'Electronics', 3, 55);

-- Sample registrations
INSERT OR IGNORE INTO registrations (student_id, course_id, academic_year,
semester_term, registration_date, grade, status)
VALUES
(1, 1, '2025-2026', 'Even', '2026-01-08', NULL, 'Enrolled'),
(1, 2, '2025-2026', 'Even', '2026-01-08', NULL, 'Enrolled'),
(1, 4, '2025-2026', 'Even', '2026-01-08', NULL, 'Enrolled'),
(2, 1, '2025-2026', 'Even', '2026-01-09', NULL, 'Enrolled'),
(2, 3, '2025-2026', 'Even', '2026-01-09', NULL, 'Enrolled'),
(3, 5, '2025-2026', 'Even', '2026-01-10', NULL, 'Enrolled'),
(4, 1, '2025-2026', 'Even', '2026-01-10', NULL, 'Enrolled'),
(4, 4, '2025-2026', 'Even', '2026-01-10', NULL, 'Enrolled');

COMMIT;
```

Query and output:

```
1          CS301        Database Management Systems  3
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$  printf "\n[Query 1] Students enrolled in course_id
 = 1\n" && sqlite3 -header -column ./aiassitantcoding/university_registration.db "SELECT s.* FROM Students s JOIN Registrations r ON
s.student_id = r.student_id WHERE r.course_id = 1;" && printf "\n[Query 2] Faculty teaching more than 2 courses\n" && sqlite3 -header
 -column ./aiassitantcoding/university_registration.db "SELECT faculty_id, COUNT(*) AS total_courses FROM Courses GROUP BY faculty_id
 HAVING COUNT(*) > 2;"

[Query 1] Students enrolled in course_id = 1
student_id  roll_no      full_name   email               phone       program     semester  enrollment_date  status  created_at
    updated_at
----------  -----------  ----------  ------------------  ----------  ----------  --------  ---------------  ------  ----------------
--  -------------------
1           SRU23CSE001  Aman Verma  aman.verma@sru.edu  9800000001  B.Tech CSE  5         2023-08-01       Active  2026-03-18 11:59:
26  2026-03-18 11:59:26
2           SRU23CSE002  Nisha Rao   nisha.rao@sru.edu   9800000002  B.Tech CSE  5         2023-08-01       Active  2026-03-18 11:59:
26  2026-03-18 11:59:26
4           SRU24CSE005  Megha Shah  megha.shah@sru.edu  9800000005  B.Tech CSE  3         2024-08-01       Active  2026-03-18 11:59:
26  2026-03-18 11:59:26

[Query 2] Faculty teaching more than 2 courses
faculty_id  total_courses
----------  -------------
1           3
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$
```

Justification

Copilot ensured **many-to-many mapping**
Queries optimized with **JOIN + GROUP BY**